Tablespace and Datafiles

- **Create a tablespace sha_tb01.**
- **Create a datafile named as shadb01.dbf in '/opt/stage/shadb01.dbf' with size 2GB**

STEP 1: First create a directory. The directory name is '/opt/stage'.

```
[oracle@localhost ~]$ su root
Password:
[root@localhost oracle]#
[root@localhost oracle]# mkdir -p /opt/stage
[root@localhost oracle]# chown -R oracle:oinstall /opt/stage
[root@localhost oracle]# chmod -R 775 /opt/stage
```

```
[oracle@localhost ~]$ cd /opt
[oracle@localhost opt]$ ls
ORCLfmap  rh  stage  VBoxGuestAdditions-7.2.6
[oracle@localhost opt]$
[oracle@localhost opt]$ ll
total 4
drwxr-xr-x. 3 root   root       22 Apr 21 16:23 ORCLfmap
drwxr-xr-x. 2 root   root        6 Aug 25  2018 rh
drwxrwxr-x. 2 oracle oinstall    6 Jun  4 16:58 stage
drwxr-xr-x. 7 root   root     4096 May  9 12:40 VBoxGuestAdditions-7.2.6
[oracle@localhost opt]$ pwd
/opt
[oracle@localhost opt]$ cd stage
[oracle@localhost stage]$ pwd
/opt/stage
```

STEP 2: Create a tablespace sha_tb01 and the datafile is '/opt/stage/shadb01.dbf' with size 2GB.

```
SQL> create tablespace sha_tb01 datafile '/opt/stage/shadb01.dbf' size 2g autoextend on;

Tablespace created.
```

```
SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024/1024 "SIZE(GB)",MAXBYTES/1024/1024/1024 "MAXSIZE",AUTOEXTENSIBLE
  2  FROM DBA_DATA_FILES WHERE TABLESPACE_NAME='SHA_TB01';

   FILE_ID TABLESPACE_NAME                FILE_NAME                                            SIZE(GB)    MAXSIZE AUT
---------- ------------------------------ -------------------------------------------------- ---------- ---------- ---
         2 SHA_TB01                       /opt/stage/shadb01.dbf                                      2 31.9999847 YES
```

- **How to resize datafile shadb01.dbf to 4GB**

```
SQL> ALTER DATABASE DATAFILE '/opt/stage/shadb01.dbf' RESIZE 4G;

Database altered.

SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024/1024 "SIZE(GB)",MAXBYTES/1024/1024/1024 "MAXSIZE",AUTOEXTENSIBLE
  2  FROM DBA_DATA_FILES WHERE TABLESPACE_NAME='SHA_TB01';

   FILE_ID TABLESPACE_NAME                FILE_NAME                                            SIZE(GB)    MAXSIZE AUT
---------- ------------------------------ -------------------------------------------------- ---------- ---------- ---
         2 SHA_TB01                       /opt/stage/shadb01.dbf                                      4 31.9999847 YES
```

➢ How to check list of datafiles and tablespaces in database.

```
SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024/1024 "SIZE(GB)",MAXBYTES/1024/1024/1024 "MAXSIZE",AUTOEXTENSIBLE
  2  FROM DBA_DATA_FILES;

   FILE_ID TABLESPACE_NAME                FILE_NAME                                            SIZE(GB)    MAXSIZE AUT
---------- ------------------------------ -------------------------------------------------- ---------- ---------- ---
         1 SYSTEM                         /u01/app/oracle/oradata/ORCL/system01.dbf            .8984375 31.9999847 YES
         3 SYSAUX                         /u01/app/oracle/oradata/ORCL/sysaux01.dbf          .712890625 31.9999847 YES
         4 UNDOTBS1                       /u01/app/oracle/oradata/ORCL/undotbs01.dbf          .33203125 31.9999847 YES
         7 USERS                          /u01/app/oracle/oradata/ORCL/users01.dbf           .016601563   .0390625 YES
         5 TX_DATA                        /u01/app/oracle/oradata/ORCL/TX_DATA01.dbf         .009765625          0 NO
         2 SHA_TB01                       /opt/stage/shadb01.dbf                                      4 31.9999847 YES

6 rows selected.
```

➢ Command to check tablespace utilization (%free,%used)

```
SQL> SELECT
    df.tablespace_name,
    ROUND(df.total_mb,2) total_mb,
    ROUND(df.total_mb - NVL(fs.free_mb,0),2) used_mb,
    ROUND(NVL(fs.free_mb,0),2) free_mb,
    ROUND((df.total_mb - NVL(fs.free_mb,0)) / df.total_mb * 100,2) used_pct
FROM
(
    SELECT
        tablespace_name,
        SUM(bytes)/1024/1024 total_mb
    FROM dba_data_files
    GROUP BY tablespace_name
) df
LEFT JOIN
(
    SELECT
        tablespace_name,
        SUM(bytes)/1024/1024 free_mb
    FROM dba_free_space
    GROUP BY tablespace_name
) fs
ON df.tablespace_name = fs.tablespace_name
ORDER BY used_pct DESC;  2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17   18   19   20   21   22   23   24

TABLESPACE_NAME                  TOTAL_MB    USED_MB    FREE_MB   USED_PCT
------------------------------ ---------- ---------- ---------- ----------
SYSTEM                                920        915          5      99.46
SYSAUX                                730     664.94      65.06      91.09
USERS                                  17      15.25       1.75      89.71
TX_DATA                                10          1          9         10
UNDOTBS1                              340      31.25     308.75       9.19
SHA_TB01                             4096          1       4095        .02

6 rows selected.
```

➢ How will you check the number of datafile & its size which belongs to spcific tablespace.

```
SQL> SELECT tablespace_name, COUNT(*) AS no_of_datafiles, ROUND(SUM(bytes)/1024/1024,2) AS total_size_mb FROM dba_data_files GROUP BY tablespace_name;

TABLESPACE_NAME                NO_OF_DATAFILES TOTAL_SIZE_MB
------------------------------ --------------- -------------
SYSTEM                                       1           920
SYSAUX                                       1           730
UNDOTBS1                                     1           340
USERS                                        1            17
TX_DATA                                      1            10
SHA_TB01                                     1          4096

6 rows selected.
```

➢ How to Drop non default datafile?

Common default/system tablespaces:

- SYSTEM
- SYSAUX
- UNDOTBS1 (or your Undo tablespace)
- TEMP
- USERS

STEP 1: Check the non-default tablespaces.

```
SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024/1024 "SIZE(GB)",MAXBYTES/1024/1024/1024 "MAXSIZE",AUTOEXTENSIBLE
  2  FROM DBA_DATA_FILES WHERE TABLESPACE_NAME NOT IN  ('SYSTEM','SYSAUX','UNDOTBS1','TEMP','USERS');

   FILE_ID TABLESPACE_NAME                FILE_NAME                                            SIZE(GB)    MAXSIZE AUT
---------- ------------------------------ -------------------------------------------------- ---------- ---------- ---
         5 TX_DATA                        /u01/app/oracle/oradata/ORCL/TX_DATA01.dbf         .009765625          0 NO
         2 SHA_TB01                       /opt/stage/shadb01.dbf                                      4 31.9999847 YES
```

```
SQL> SELECT tablespace_name FROM dba_tablespaces WHERE tablespace_name NOT IN  ('SYSTEM','SYSAUX','UNDOTBS1','TEMP','USERS');

TABLESPACE_NAME
------------------------------
TX_DATA
SHA_TB01
```

STEP 2: Drop these two tablespaces

```
SQL> DROP TABLESPACE TX_DATA INCLUDING CONTENTS AND DATAFILES;

Tablespace dropped.

SQL> DROP TABLESPACE SHA_TB01 INCLUDING CONTENTS AND DATAFILES;

Tablespace dropped.

SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,BYTES/1024/1024/1024 "SIZE(GB)",MAXBYTES/1024/1024/1024 "MAXSIZE",AUTOEXTENSIBLE
  2  FROM DBA_DATA_FILES WHERE TABLESPACE_NAME NOT IN  ('SYSTEM','SYSAUX','UNDOTBS1','TEMP','USERS');

no rows selected
```